

Lab Exercise 13

SIMPLE PROGRAMMING USING CASE AND LOOP

AIM

To implement **CASE** and **LOOP with ITERATE** statements in MySQL using the VegetableDB database.

PROCEDURE

1. Select the VegetableDB database.
 2. Create a function using the **CASE** statement to classify a vegetable price.
 3. Execute the function and display the price category.
 4. Create a function using **LOOP and ITERATE** to increase a value until it reaches 4000.
 5. Execute the function and display the final value.
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Program 1: CASE Statement

Here, we use monthly_value as the **vegetable sales value**.

```
USE VegetableDB;

DELIMITER //

CREATE FUNCTION VegetablePriceLevel(monthly_value INT)
RETURNS VARCHAR(30)
DETERMINISTIC
BEGIN
    DECLARE income_level VARCHAR(30);

    CASE
        WHEN monthly_value <= 4000 THEN
            SET income_level = 'Low Value';

        WHEN monthly_value <= 5000 THEN
            SET income_level = 'Average Value';

        ELSE
            SET income_level = 'High Value';
    END CASE;

    RETURN income_level;
END //

DELIMITER ;
```

Execute

```
SELECT VegetablePriceLevel(3500);
```

```
mysql> SELECT VegetablePriceLevel(3500);
+-----+
| VegetablePriceLevel(3500) |
+-----+
| Low Value                  |
+-----+
1 row in set (0.00 sec)
```

```
SELECT VegetablePriceLevel(4500);
```

```
mysql> SELECT VegetablePriceLevel(4500);
+-----+
| VegetablePriceLevel(4500) |
+-----+
| Average Value             |
+-----+
1 row in set (0.00 sec)
```

```
SELECT VegetablePriceLevel(6000);
```

```
mysql> SELECT VegetablePriceLevel(6000);
+-----+
| VegetablePriceLevel(6000) |
+-----+
| High Value                |
+-----+
1 row in set (0.00 sec)
```

Program 2: LOOP with ITERATE

Here, the value starts at **1000** and increases by **500** until it reaches 4000.

```
USE VegetableDB;
```

```
DELIMITER //
```

```
CREATE FUNCTION VegetableLoop()
RETURNS INT
DETERMINISTIC
BEGIN
    DECLARE income INT DEFAULT 1000;

    my_loop: LOOP

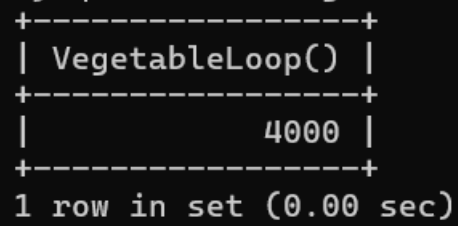
        IF income >= 4000 THEN
            LEAVE my_loop;
        END IF;

        SET income = income + 500;
```

```
        ITERATE my_loop;  
    END LOOP;  
  
    RETURN income;  
END //  
  
DELIMITER ;
```

Execute

```
SELECT VegetableLoop();
```



```
+-----+  
| VegetableLoop() |  
+-----+  
|           4000 |  
+-----+  
1 row in set (0.00 sec)
```